higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE (VOCATIONAL)

## MATHEMATICS
(First Paper)
**NQF LEVEL 3**

**4 NOVEMBER 2019**

**This marking guideline consists of 11 pages.**

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✓=1 mark √=half mark

## QUESTION 1

1.1 1.1.1 $i^{\sqrt{64}} \cdot i^8 = (i^2)^4 = 1$

1.1.2 $(2i)^5 \times i^9 = 32i^{14} = 32(i^2)^7 = -32$

$(2 \times 2)$ (4)

1.2 1.2.1 $= 12{,}6 + j\,4{,}9 - 1{,}3 - j\,9{,}6 = 11{,}3 - j\,4{,}7$ (2)

1.2.2 $(6 - i\,7)(6 + i\,7) = 36 - 49i^2 = 36 + 49 = 85$ (2)

1.2.3 $\dfrac{-7 - i2}{-4 + i} \times \dfrac{-4 - i}{-4 - i} \cdot \dfrac{28 + 7i + 8i + 2i^2}{16 - i^2} \cdot \dfrac{28 + 15i - 2}{17}$

$\cdot \dfrac{26}{17} + \dfrac{15}{17}i$ or $\dfrac{26 + i15}{17}$ (3)

1.2.4 $3(\cos 25° + i \sin 25°).\ 8(\cos 200° + i\ \sin 200°)$
$= 24\ (\cos 225° + i \sin 225°)$ (2)

1.2.5 $\dfrac{4(\cos 190° + i\ \sin 190°)}{2(\cos 70° + i \sin 70°)} = 2(\cos 120° + i \sin 120°)$ (2)

1.3 1.3.1 $z = -6 + i9 \quad r = \sqrt{36 + 81} \quad r = 3\sqrt{13} = 10{,}817 \quad \theta = \tan^{-1} - \dfrac{9}{6} \quad ref = 56{,}31°$

$\theta = 123{,}69° \quad 10{,}817 \angle \underline{123{,}69°}$ (3)

1.3.2 $z = \dfrac{-7 - i8}{2 - 3i} \times \dfrac{2 + 3i}{2 + 3i} \cdot \dfrac{10 - 37i}{13}$

$\cdot \dfrac{10}{13} - \dfrac{37}{13}i \quad r = \sqrt{0{,}592 + 8{,}101}$

$= 2{,}948 \quad ref = \tan^{-1} \dfrac{2{,}846}{0{,}769} = 74{,}878°$

$\theta = 285{,}124°$

$z = 2{,}948 \angle \underline{285{,}124°}$

Alternate

$z = -7 - i8 = \sqrt{113}\ cis 228{,}814°$ ✓

$z = 2 - 3i = \sqrt{13}\ cis 303{,}690°$ ✓

$z = \dfrac{-7 - i8}{2 - 3i} = \dfrac{\sqrt{113}\ cis 228{,}814}{\sqrt{13}\ cis 303{,}690°}$

✓ ✓

$z = 2{,}948\ cis(-74{,}876°)$

$z = 2{,}948\ cis 285{,}124°$ ✓ (5)

1.4 1.4.1 $\dfrac{12 \angle \underline{45°} \,.\, 6 \angle \underline{180°}}{8 \angle \underline{20°}} \cdot 72 \angle \underline{225°} \cdot \cdot \cdot = 9 \angle \underline{205°} \cdot \cdot = -8{,}157 - i\,3{,}804$ (4)

1.4.2 $\cdot \dfrac{3{,}8(\cos 120° + i \sin 120°)}{2{,}5 \cos 45°}$ (3)

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✓ ✓

$= 1{,}52 \angle 75° ¿¿$

$= 0{,}393 + i\,1{,}468$

**[30]**

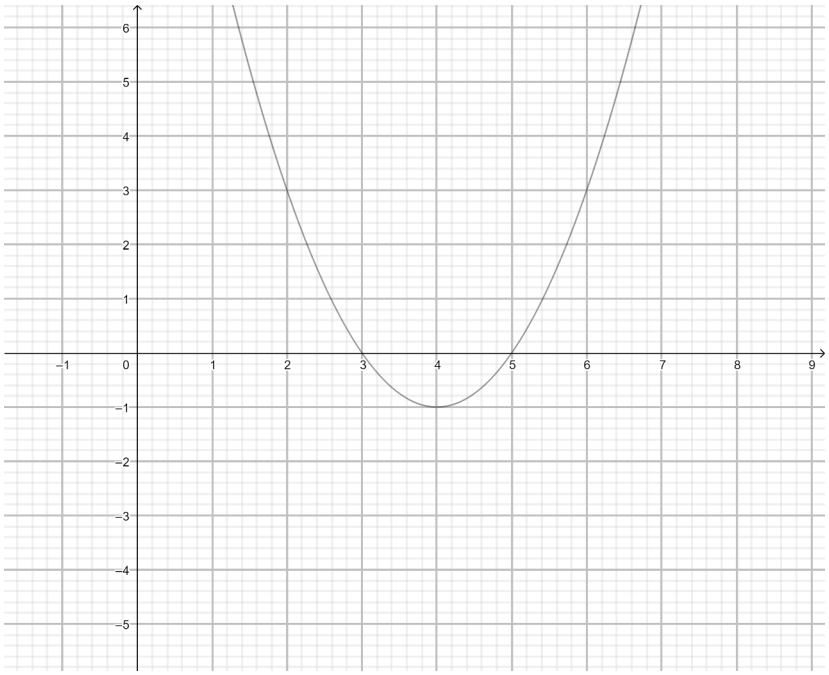
## QUESTION 2

2.1 $x^2-8x+23\leq 8x^2-8x+15\leq 0(x-3)(x-5)$

| | $x\leq 3$ | $3\leq x\leq 5$ | $x\geq 5$ |
|---|---|---|---|
| $x-3$ | $-$¿ | $+$¿ | $+$¿ |
| $x-5$ | $-$¿ | $-$¿ | $+$¿ |
| $(x-3)(x-5)$ | $+$¿ | $-$¿ | $+$¿ |

$\therefore 3\leq x\leq 5$

The answer could be found by drawing the parabola $y=x^2-8x+15$




Marks should be awarded as follows
✓ for drawing a parabola
✓✓ one mark for each correct root
✓ for a TP
$\therefore 3\leq x\leq 5$ ✓

(5)

2.2 $5x^2-3=2x$

$x^2-\frac{2}{5}x=\frac{3}{5}$

$x^2-\frac{2}{5}x+(\frac{2}{10})^2=\frac{3}{5}+(\frac{2}{10})^2$

$(x-\frac{1}{5})^2=\frac{3}{5}+\frac{1}{25}(x-\frac{1}{5})^2=\frac{16}{25}x-\frac{1}{5}=\pm\frac{4}{5}$

$\therefore x=-\frac{3}{5}$ or $x=1$

**OR**

$(5x^2-2x)-3=05\left(x^2-\frac{2}{5}x\right)-3=0$

$5\left(x^2-\frac{2}{5}x+\frac{1}{25}\right)-3-5\left(\frac{1}{25}\right)=0$

$5(x-\frac{1}{5})^2=\frac{16}{5}(x-\frac{1}{5})^2=\frac{16}{25}$

(4)

2.3 $y+2x=2$ *equation* $1\,y^2+2x^2=3xy$ *equation* $2¿$ *eq*.$1: y=2-2x$

*substitute* $\in$ *eq*. $2(2-2x)^2+2x^2=3(2-2x)x4-8x+4x^2+2x^2-6x+6x^2=0$

$12x^2-14x+4=0\ (\div 2)6x^2-7x+2=0(3x-2)(2x-1)=0x=\frac{2}{3}$ or $x=\frac{1}{2}$ *substitute* $x$:

$x=\frac{2}{3}\quad y=2-2\left(\frac{2}{3}\right)y=\frac{2}{3}x=\frac{1}{2}\quad y=2-2\left(\frac{1}{2}\right)y=1$

(4)

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2.4 2.4.1 $\frac{a^2-a}{a-2}\cdot\frac{a+1}{a^2+4a}\div\frac{a^2-3a-4}{a^2-16}$

¿ $\frac{a(a-1)}{a-2}\cdot\frac{a+1}{a(a+4)}\times\frac{(a-4)(a+4)}{(a+1)(a-4)}$

¿ $\frac{a-1}{a-2}$ (4)

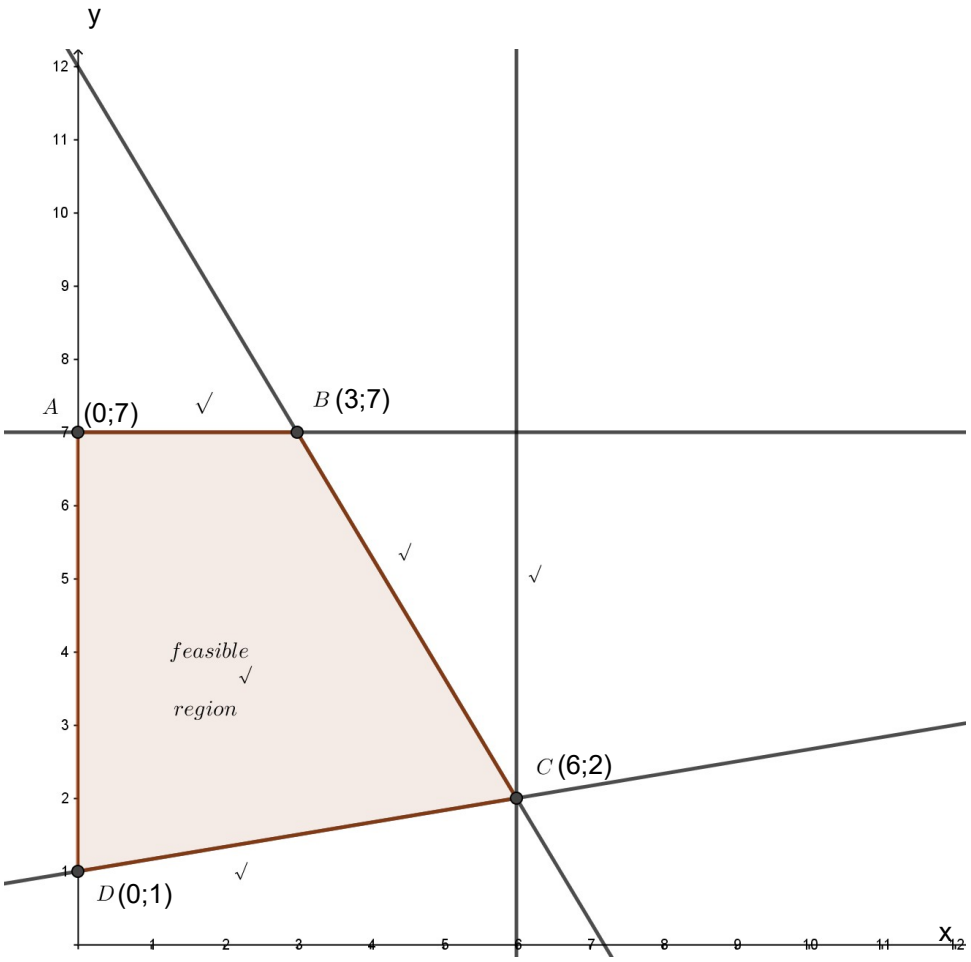
2.4.2 $\frac{y}{y^2-16}-\frac{y+1}{y^2-5y+4}$ ¿ $\frac{y}{(y-4)(y+4)}-\frac{y+1}{(y-1)(y-4)}$

¿ $\frac{y(y-1)-(y+1)(y+4)}{(y-1)(y-4)(y+4)}$ ¿ $\frac{y^2-y-y^2-5y-4}{(y-1)(y-4)(y+4)}$ ¿ $\frac{-6y-4}{(y-1)(y-4)(y+4)}$

¿ $\frac{-2(3y+2)}{(y-1)(y-4)(y+4)}$

(4)

2.5 2.5.1 & 2.5.2




(5)

2.5.3 $P = 100x + 200y$

A(0;7)$\therefore P = 100(0) + 200(7) = 1\ \ 400$ √

B(3;7)$\therefore P = 100(3) + 200(7) = 1\ \ 700$

C(6;2)$\therefore P = 100(6) + 200(2) = 1\ \ 000$

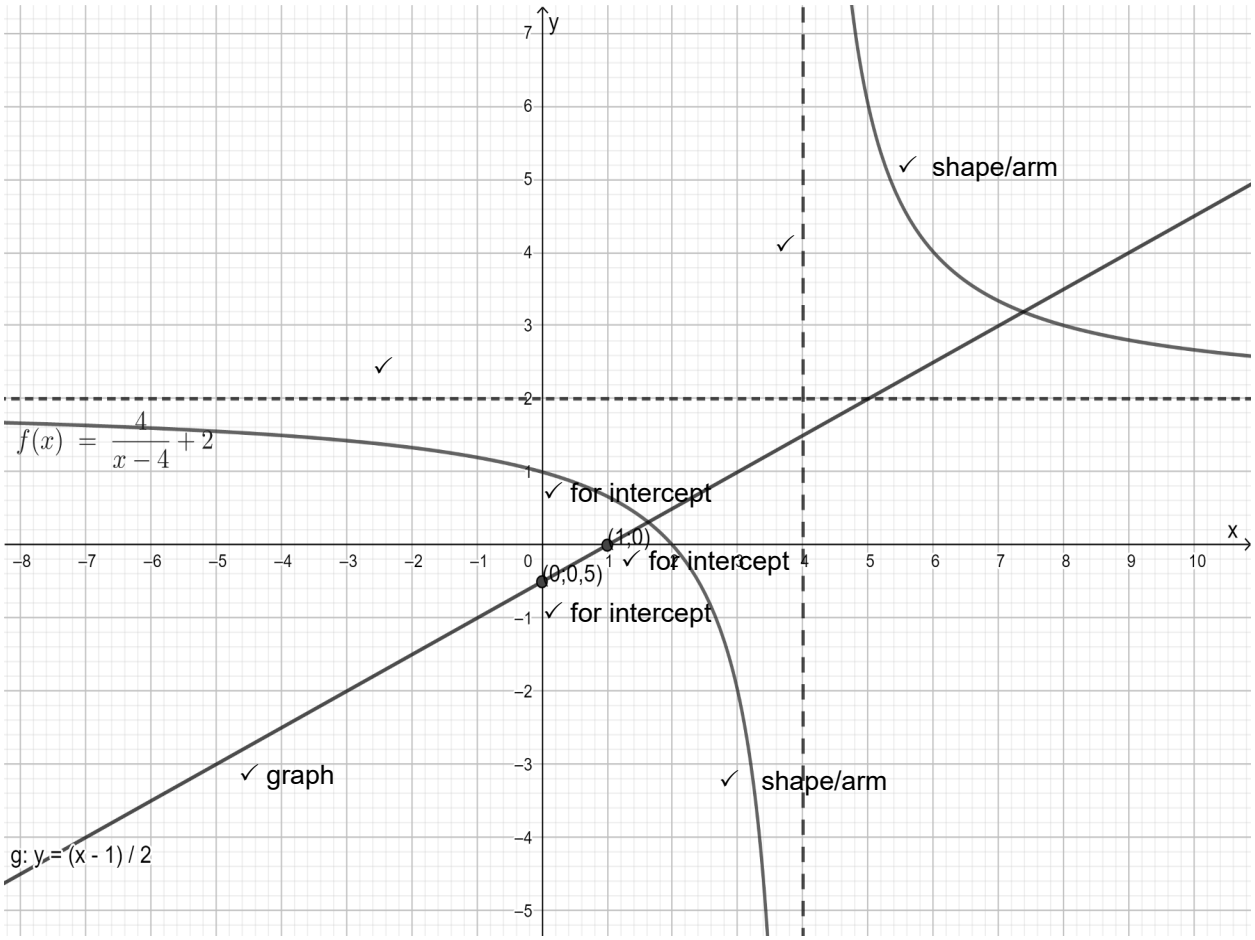
D(0;1)$\therefore P = 100(0) + 200(1) = 2\ \ 000$ √

Maximum profit at B(3;7)✓ (4)

**[30]**

## QUESTION 3

3.1




(8)

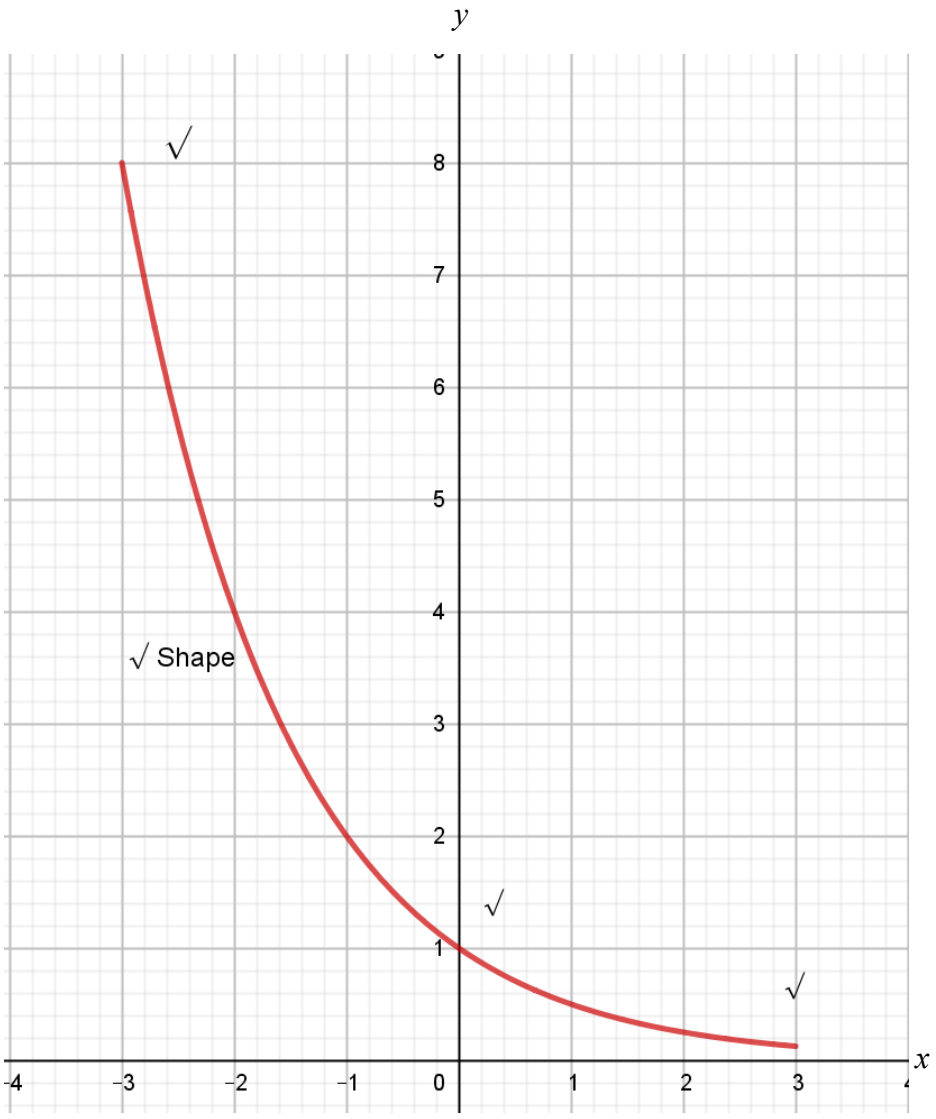
3.2 3.2.1 $y = a(x - x_1)(x - x_2)$ $y = a(x - (1))(x - (3))$ $y = a(x^2 - 4x + 3)$ $(0,3)\ 3 = 3a\ a = 1$

$y = x^2 - 4x + 3$

3.2.2 $\text{T/P} :-\frac{b}{2a}=-\frac{(-4)}{2(1)}x=2y=(2)^2-4(2)+3y=-1(2;-1)$

(2 × 4) (8)

3.3




(4)
**[20]**

## QUESTION 4

4.1 $s(t) = t^2+3ts(t+h)=(t+h)^2+3(t+h)¿t^2+2th+h^2+3t+3h$

$s'=\lim_{h\to 0}\frac{t^2+2th+h^2+3t+3h-t^2-3t}{h}¿\lim_{h\to 0}\frac{2th+h^2+3h}{h}\ ¿\lim_{h\to 0}\frac{h(2t+h+3)}{h}$

$¿\lim_{h\to 0}2t+h+3=2t+0+3=2t+3$ (4)

4.2 $\lim ¿_{x\to\infty}\frac{4-x^2}{x^3-2}¿$

$$=\lim_{x\to\infty}\frac{\frac{4}{x^3}-\frac{x^2}{x^3}}{\frac{x^3}{x^3}-\frac{2}{x^3}}$$

$$¿\lim ¿_{x\to\infty}\frac{\frac{4}{x^3}-\frac{1}{x}}{1-\frac{2}{x^3}}¿¿\frac{\frac{4}{\infty}-\frac{1}{\infty}}{1-\frac{2}{\infty}}\ ¿\frac{0-0}{1-0}=0$$

(3)

4.3 4.3.1 $y=\frac{1}{4}x^8-\frac{1}{2}x^4+2y'=2x^7-2x^3+0$

4.3.2 $y=\frac{2}{x}-\sqrt{x}$

$=2x^{-1}-x^{\frac{1}{2}}$ ✓

$=-2x^{-2}-\frac{1}{2}x^{-\frac{1}{2}}$ ✓

$=-\frac{2}{x^2}-\frac{1}{2\sqrt{x}}$ ✓

4.3.3 $y=(x+2)(3x-5)y=3x^2-5x+6x-10\ \ y=3x^2+x-10y'=6x+1$

(3 × 3) (9)

4.4 $P=x(32-x)P=32x-x^2P'=32-2x32-2x=0-2x=-32x=16\ \ \therefore P=16(32-16)$

$P=256$ (4)

**[20]**

**TOTAL: 100**